

Problem 1:

K of $\mathbb{R}^2$ orthonormal coord. sys.

$A(4, -1), B(3, 6), C(-5, 2) \rightarrow \triangle ABC$.

K' coord. sys. w origin A and basis $B' = (\vec{AB}, \vec{AC})$.

a) $\triangle ABC \rightarrow$ right-angled?

$$\vec{AB} = B - A = (-1, 7)$$

$$\vec{AC} = C - A = (-9, 3)$$

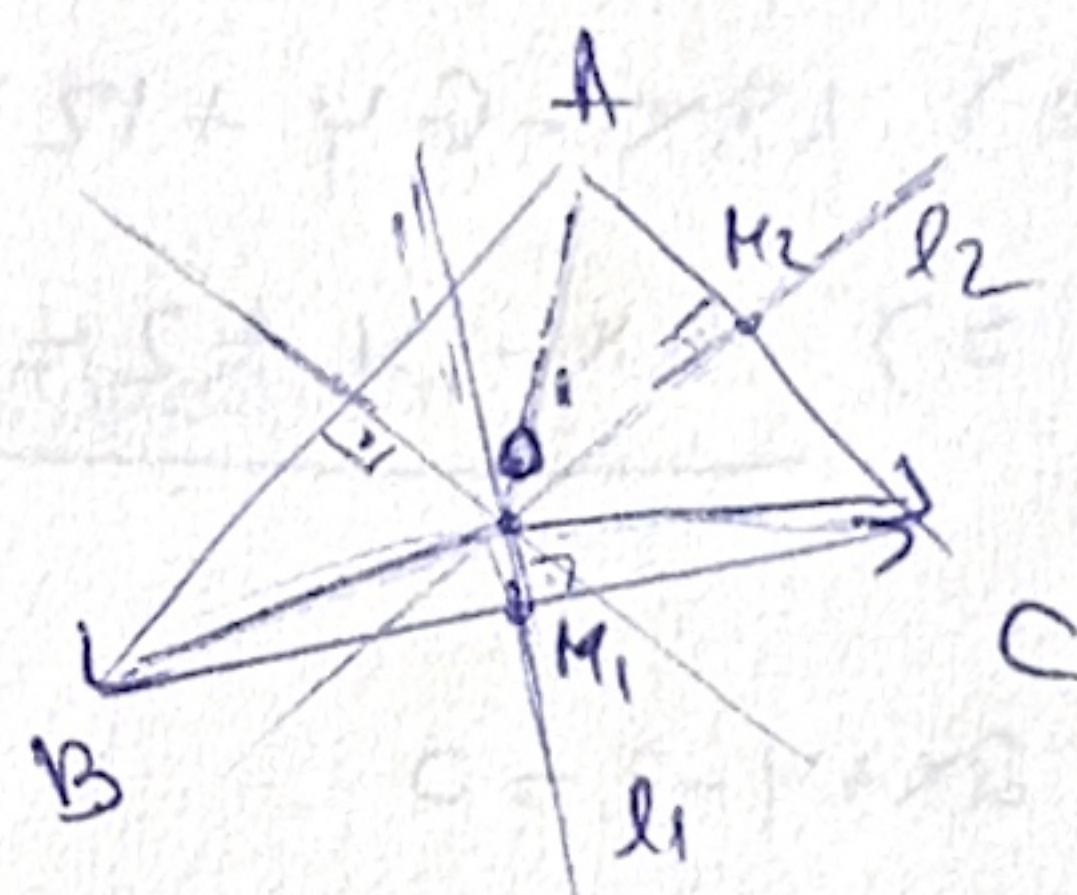
$$\vec{BC} = C - B = (-8, -4)$$

$$\vec{AB} \cdot \vec{AC} = (-1, 7) \cdot (-9, 3) = -1 \cdot (-9) + 7 \cdot 3 = 9 + 21 = 30 \neq 0$$

$$\vec{AB} \cdot \vec{BC} = (-1, 7) \cdot (-8, -4) = 8 - 28 = -20 \neq 0$$

$$\vec{AC} \cdot \vec{BC} = (-9, 3) \cdot (-8, -4) = 72 - 12 = 60 \neq 0.$$

$\Rightarrow$ no dot product $= 0 \Rightarrow \triangle ABC$ is not r-angled.



b) Circumcenter (O') of $\triangle ABC$ w.r. to K . $\rightarrow O'(x, y)$.

let l_1 mediator of BC ; H_1 - midpoint of BC

l_2 mediator of AC ; H_2 - midpoint of AC .

$$\vec{OH_1} = \frac{\vec{OB} + \vec{OC}}{2} \Rightarrow H_1 \left(\frac{x_B + x_C}{2}, \frac{y_B + y_C}{2} \right)$$

$$\Rightarrow H_1 \left(\frac{3 + (-5)}{2}, \frac{6 + 2}{2} \right) \Rightarrow H_1 \left(-1, 4 \right)$$

$$\vec{OH_2} = \frac{\vec{OA} + \vec{OC}}{2} \Rightarrow H_2 \left(\frac{x_A + x_C}{2}, \frac{y_A + y_C}{2} \right)$$

$$\Rightarrow H_2 \left(\frac{-1 + (-5)}{2}, \frac{-1 + 2}{2} \right) \Rightarrow H_2 \left(-3, \frac{1}{2} \right)$$

$O'H_1 \perp BC$

$$\Rightarrow \angle \vec{O'H_1}, \vec{BC} = 0 \Rightarrow \langle (-1-x, 4-y), (-8, -4) \rangle = 0$$

$$\Rightarrow \langle (-1-x, 4-y), (-8, -4) \rangle = 0$$

$$\Rightarrow \langle (-1-x, 4-y), (-8, -4) \rangle = 0$$

$$\Rightarrow -8(-1-x) - 4(4-y) = 0 \Rightarrow 8 + 8x - 16 + 4y = 0$$

$$\Rightarrow 8x + 4y - 8 = 0 \quad | :4 \Rightarrow 2x + y - 2 = 0.$$

$$O'H_2 \perp AC \Rightarrow \langle \vec{O'H_2}, \vec{AC} \rangle = 0$$

$$\Rightarrow \langle (-\frac{1}{2}-x, \frac{1}{2}-y), (-3, 3) \rangle = 0$$

$$\Rightarrow -9(-\frac{1}{2}-x) + 3(\frac{1}{2}-y) = 0 \quad | \cdot 2$$

$$\Rightarrow -9(-1-2x) + 3(1-2y) = 0$$

$$\Rightarrow 9 + 18x + 3 - 6y = 0$$

$$\Rightarrow 18x - 6y + 12 = 0 \quad | :6$$

$$\Rightarrow \underline{3x - y + 2 = 0}$$

$$\begin{cases} 2x + y - 2 = 0 \\ 3x - y + 2 = 0 \end{cases} \quad (+)$$

$$5x = 0 \Rightarrow x = 0$$

$$\Rightarrow y = 2$$

$$\Rightarrow O'(0, 2)$$

c) w.r. to K' ,

median of $\triangle ABC$ passing through B.

$K': (A, \vec{AB}, \vec{AC}); A\text{-origin} \Rightarrow A(0,0)$.

$$\vec{AB} = A + u \cdot \vec{AB} + v \cdot \vec{AC} \Rightarrow u = 1, v = 0$$

$$\Rightarrow \vec{AB}(1, 0)$$

$$\vec{AC} = A + u \cdot \vec{AB} + v \cdot \vec{AC} \Rightarrow u = 0; v = 1$$

$$\Rightarrow \vec{AC}(0, 1)$$

H - midpoint of AC.

$$H = \frac{A+C}{2} = \frac{(0,0) + (-5,2)}{2} = \frac{(-5,2)}{2} = \left(-\frac{5}{2}, 1\right)$$

$$= \frac{(0,0) + (0,1)}{2} = \frac{(0,1)}{2} = \left(0, \frac{1}{2}\right)$$

(2D): line passing through B and H:

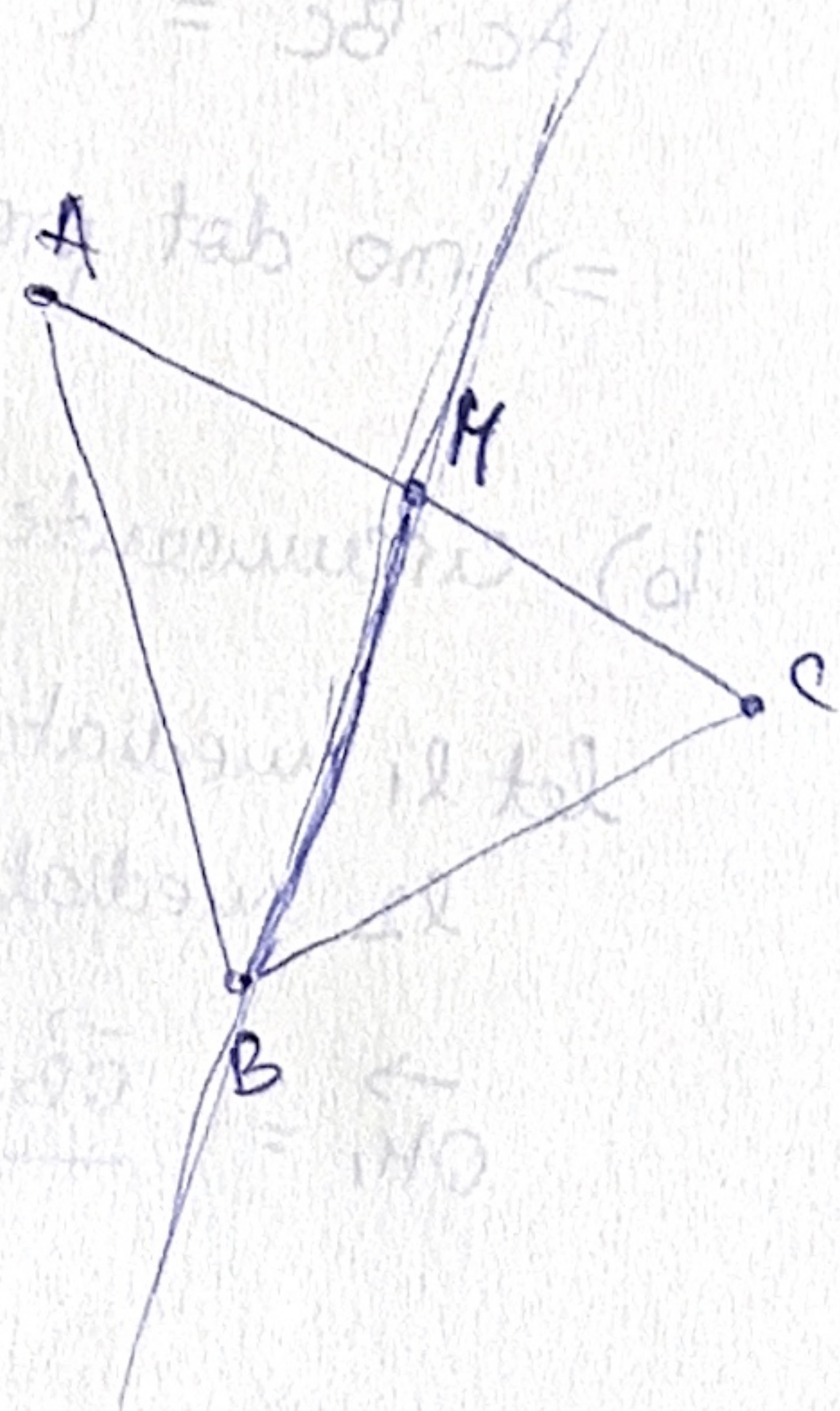
$$\begin{vmatrix} x & y & 1 \\ 0 & \frac{1}{2} & 1 \\ 1 & 0 & 1 \end{vmatrix} = 0 \Rightarrow x \cdot \begin{vmatrix} \frac{1}{2} & 1 \\ 0 & 1 \end{vmatrix} - y \cdot \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} + 1 \cdot \begin{vmatrix} 0 & \frac{1}{2} \\ 1 & 0 \end{vmatrix} =$$

$$= x \left(\frac{1}{2} - 0\right) - y(-1) + 1 \cdot (0 - \frac{1}{2})$$

$$= \frac{1}{2}x + y - \frac{1}{2} = 0 \quad | \cdot 2 \Rightarrow \underline{x + 2y - 1 = 0}$$

median line

(2)



$$1) A_{\Delta ABC} = \frac{1}{2} \cdot |\det| = \frac{1}{2} |60| = \frac{60}{2} = 30.$$

$$\det = \begin{vmatrix} x_A & y_A & 1 \\ x_B & y_B & 1 \\ x_C & y_C & 1 \end{vmatrix} = \begin{vmatrix} 4 & -1 & 1 \\ 3 & 6 & 1 \\ -5 & 2 & 1 \end{vmatrix} = \underline{24 + 6 + 5 + 3 - 8 + 30} = 60.$$

Problem 2.

4-or. orthonormal sys. of $\mathbb{E}^3$

A(1,2,0), B(1,2,1), C(0,3,2), D(1,1,1) tetrahedron.

$K' : (B, \vec{AB}, \vec{AC}, \vec{AD})$.

$$? a) \vec{AB} = B - A = (1,2,1) - (1,2,0) = (0,0,1).$$

$$\vec{AC} = C - A = (-1,1,2).$$

$$\vec{AD} = D - A = (0,-1,1)$$

$$\Rightarrow H = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 1 & -1 \\ \textcircled{1} & 2 & 1 \end{bmatrix} \Rightarrow \det H = \begin{vmatrix} 0 & -1 & 0 \\ 0 & 1 & -1 \\ \textcircled{1} & 2 & 1 \end{vmatrix} = 1 \cdot (-1)^4 \cdot \begin{vmatrix} -1 & 0 \\ 1 & -1 \end{vmatrix} = 1.$$

$\vec{AB} \quad \vec{AC} \quad \vec{AD}$

$\Rightarrow$ right oriented (like B).

b) Gram-Schmidt $\rightarrow$ transform B' in an orthogonal basis.

$B' = (\vec{AB}, \vec{AC}, \vec{AD})$. right-oriented basis.

$K' = (B, B')$ non-orthonormal frame.

$B' = \left(\underset{e_1}{(0,0,1)}, \underset{e_2}{(-1,1,2)}, \underset{e_3}{(0,-1,1)} \right)$ basis of $\mathbb{V}^3 / \mathbb{E}^3 ? / \mathbb{A}^3 ?$

1) Construct an orthogonal basis: e_1', e_2', e_3' .

$$e_1' = e_1$$

$$e_2' = e_2 - \frac{\langle e_1', e_2 \rangle}{\langle e_1', e_1' \rangle} \cdot e_1' = e_2 - \text{Pr}_{e_1'}^\perp(e_2)$$

$$(-1,1,2) -$$

$$\langle e_1', e_2 \rangle = \langle (0,0,1), (-1,1,2) \rangle = 0 \cdot (-1) + 0 \cdot 1 + 1 \cdot 2 = 2.$$

$$\langle e_1', e_1' \rangle = \langle (0,0,1), (0,0,1) \rangle = 1.$$

$$= (-1,1,2) - \frac{2}{1} \cdot (0,0,1) = (-1,1,2) - (0,0,2) = (-1,1,0). \textcircled{3}$$

$$e_3' = e_3 - \frac{\langle e_1', e_3 \rangle}{\langle e_1', e_1' \rangle} \cdot e_1' - \frac{\langle e_2', e_3 \rangle}{\langle e_2', e_2' \rangle} \cdot e_2' = e_3 - \text{Pr}_{e_1'}^\perp(e_3) - \text{Pr}_{e_2'}^\perp(e_3)$$

$$\langle e_1', e_3 \rangle = \langle (0, 0, 1), (0, -1, 1) \rangle = 1$$

$$\langle e_1', e_1' \rangle = 1.$$

$$\langle e_2', e_3 \rangle = \langle (-1, 1, 0), (0, -1, 1) \rangle = -1.$$

$$\langle e_2', e_2' \rangle = \langle (-1, 1, 0), (-1, 1, 0) \rangle = 1 + 1 = 2.$$

$$= (0, -1, 1) - \frac{1}{1} \cdot (0, 0, 1) - \frac{-1}{2} (-1, 1, 0)$$

$$= (0, -1, 1) - (0, 0, 1) + \frac{1}{2} (-1, 1, 0)$$

$$= (0, -1, 0) + \left(-\frac{1}{2}, \frac{1}{2}, 0\right)$$

$$= \left(-\frac{1}{2}, -\frac{1}{2}, 0\right) \cdot 1 \cdot 2$$

$$\Rightarrow e_3' = (-1, -1, 0); |e_3'| = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}.$$

$$\Rightarrow e_2' = (-1, 1, 0); |e_2'| = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}$$

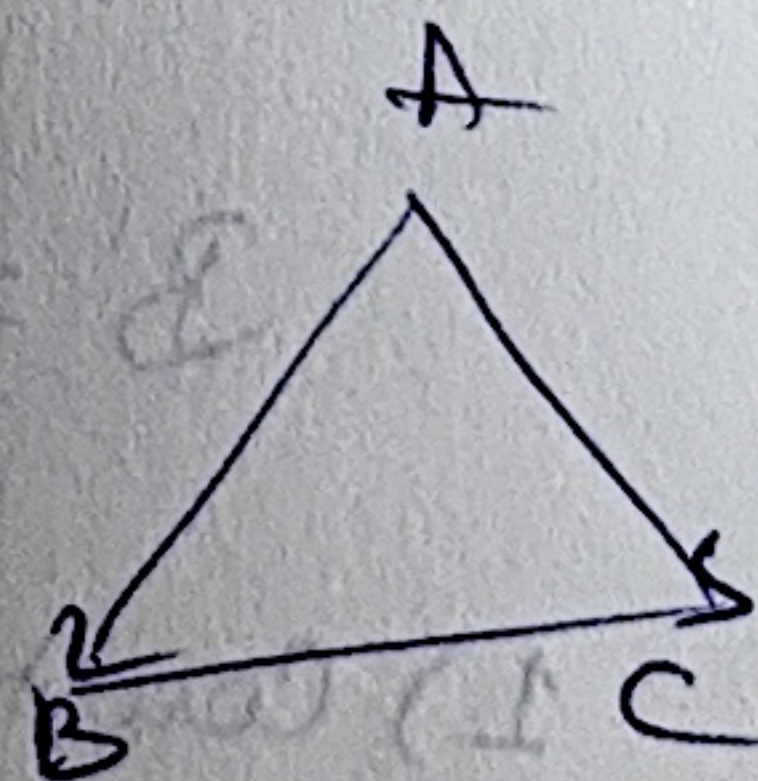
$$\Rightarrow e_1' = (0, 0, 1); |e_1'| = \sqrt{0^2 + 0^2 + 1^2} = 1.$$

2) Normalize:

$$B'' = \left\{ \frac{e_1'}{|e_1'|}, \frac{e_2'}{|e_2'|}, \frac{e_3'}{|e_3'|} \right\}$$

$$= \left\{ \frac{(0, 0, 1)}{1}, \frac{(-1, 1, 0)}{\sqrt{2}}, \frac{(-1, -1, 0)}{\sqrt{2}} \right\}$$

$$= \left\{ (0, 0, 1), \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right), \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right) \right\}.$$



c) orthogonal projection of D on plane containing $\triangle ABC$.

$$\vec{AB} = B - A = (0, 0, 1)$$

$$\vec{AC} = C - A = (-1, 1, 2)$$

$$m = \vec{AB} \times \vec{AC} = \begin{vmatrix} i & j & k \\ 0 & 0 & 1 \\ -1 & 1 & 2 \end{vmatrix} = 1 \cdot (-1) \cdot \begin{vmatrix} i & j \\ -1 & 1 \end{vmatrix}$$

$$= -(i + j) = -i - j = (-1, -1, 0)$$

$$\Rightarrow m = (1, 1, 0) \text{ normal vector.}$$

$$A(1,2,0), m = (1,1,0):$$

$$(ABC): 1 \cdot (x - x_A) + 1 \cdot (y - y_A) + 0 \cdot (z - z_A) = 0$$

$$(x - 1) + 1 \cdot (y - 2) = 0$$

$$\underline{x + y - 3 = 0 \quad (1)}$$

$$(x, y, z) = D + t \cdot m$$

$$= (1, 1, 1) + t \cdot (1, 1, 0)$$

$$= (1+t, 1+t, 0)$$

$$\Rightarrow \begin{cases} x = 1+t \\ y = 1+t \\ z = 0 \end{cases} \quad (2)$$

$$(1), (2) \Rightarrow (1+t) + (1+t) - 3 = 0$$

$$\Rightarrow 2t + 2 - 3 = 0$$

$$\Rightarrow 2t - 1 = 0$$

$$\Rightarrow t = \frac{1}{2}$$

$$\Rightarrow P(x, y, z) = \left(1 + \frac{1}{2}, 1 + \frac{1}{2}, 0\right) = \left(\frac{3}{2}, \frac{3}{2}, 0\right)$$

projection
point

d) w.r. to K' give eq. of the plane containing ^{face} BCD of tetrahedron.

$$K': (B, \vec{AB}, \vec{AC}, \vec{AD}).$$

$$\underline{B_{K'} = (0, 0, 0)}.$$

$$\vec{Bx} = x' \cdot \vec{AB} + y' \cdot \vec{AC} + z' \cdot \vec{AD}$$

$$\vec{Bc} = c - B = (c - A) - (B - A) = \vec{AC} - \vec{AB} \Rightarrow x' = -1$$

$$\Rightarrow y' = 1$$

$$\Rightarrow z' = 0.$$

$$\Rightarrow \underline{C_{K'} = (-1, 1, 0)}.$$

$$\vec{Bd} = D - B = (D - A) - (B - A) = \vec{AD} - \vec{AB} \Rightarrow x' = -1$$

$$\Rightarrow y' = 0$$

$$\Rightarrow z' = 1.$$

$$\Rightarrow \underline{D_{K'} = (-1, 0, 1)}$$

$$(BCD): ax' + b \cdot y' + c \cdot z' = 0$$

$$C' \in (BCD) \Rightarrow a \cdot (-1) + b \cdot 1 + c \cdot 0 = 0 \Rightarrow b - a = 0 \Rightarrow \underline{a = b}.$$

$$D' \in (BCD) \Rightarrow a \cdot (-1) + b \cdot 0 + c \cdot 1 = 0 \Rightarrow -a + c = 0 \Rightarrow \underline{a = c}$$

(5)

$$(BCD): a \cdot x' + a \cdot y' + a \cdot z' = 0 \quad | : a$$

$$\underline{x' + y' + z' = 0.}$$

Pb3: $a, b, c, d \in \mathbb{R}^3.$

$$(a \times b) \times (c \times d) \stackrel{?}{=} [a, c, d]b - [b, c, d]a$$

Grassmann's.

Grassmann's identity: (lin. alg.)

$$\boxed{\dim(U+V) = \dim(U) + \dim(V) - \dim(U \cap V)} \quad ?$$

$$\underset{z}{x} \times (\underset{z}{y} \times \underset{z}{z}) = (x \cdot z)y - (x \cdot y)z$$

$$(a \times b) \times (c \times d) = (\underbrace{(a \times b) \cdot d}_{[a, b, d]})c - (\underbrace{(a \times b) \cdot c}_{[a, b, c]})d$$

$$= [a, b, d]c - [a, b, c]d$$

$$= [a, c, d]b - [b, c, d]a$$

prop. of
determinants
←

Pb4: Distance formula from a point to a hyperplane in $\mathbb{E}^n$.

$K \rightarrow$ frame of $\mathbb{E}^n$.

$P(p_1, \dots, p_m) \rightarrow$ point

$H: \underline{a}_1 x_1 + \dots + \underline{a}_m x_m = b \rightarrow$ hyperplane.

$$d(P, H) = \frac{|a_1 \cdot p_1 + a_2 \cdot p_2 + \dots + a_m \cdot p_m - b|}{\sqrt{a_1^2 + a_2^2 + \dots + a_m^2}}$$

Proof:

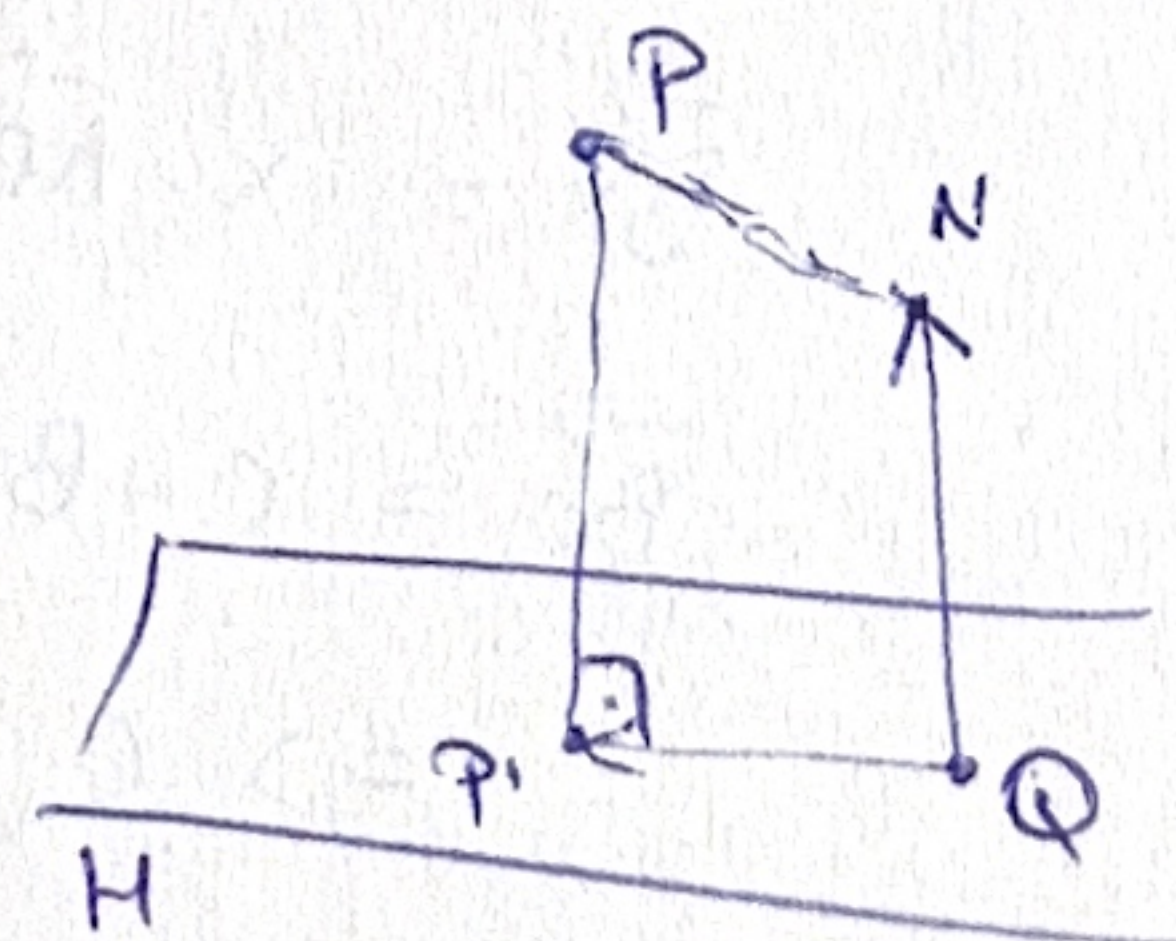
$\perp$ from P to $H \Rightarrow \cap H$ in P' .

$$\Rightarrow d(P, H) = |PP'|.$$

$Q \in H$

$Q \neq P'$

$m = m(a_1, \dots, a_m)$ normal vector.



how does it look?

$\Rightarrow |PP'| =$ length of orthogonal projection of $\vec{QP}$ onto m .

let N s.t. $m = \vec{QN}$

$$d(P, H) = \left| \frac{\langle m, \vec{QP} \rangle}{|m|^2} \cdot m \right| = \frac{|\langle m, \vec{QP} \rangle|}{|m|} = \frac{|\langle m, \vec{OP} - \vec{OQ} \rangle|}{|m|} =$$

$$= \frac{|\langle m, \vec{OP} \rangle - b|}{|m|}$$